

Monte Carlo Simulation for Tail Risk

Calibration, Simulation, and the Gap with Reality

venkat

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Abstract

These are working notes on understanding *tail risk* through Monte Carlo simulation. The guiding question is the relationship between **calibration** (fitting a model to data we have seen) and **reality** (the losses the world actually delivers). I build a dataset from a known fat-tailed, volatility-clustering process, calibrate three risk models to it (Normal, Student- t , and an Extreme-Value/Generalized-Pareto tail model), estimate Value-at-Risk and Expected Shortfall by Monte Carlo, and then *backtest* each calibration against the realised series. The recurring lesson: a model can fit the body of the data and still be dangerously wrong in the tail, and calibration only ever recovers the model you assumed — not necessarily the process that generated the world.

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1 The question

A risk number is a claim about the future stated as a probability. When we say “the one-day 99% Value-at-Risk is 3.2%,” we are asserting that a loss worse than 3.2% should happen on about one day in a hundred. Two separate things must go right for that claim to hold:

1. **Calibration** — we picked model parameters that match the data.
2. **Specification** — the model *family* is capable of describing reality in the first place.

Calibration is an optimisation problem and always “succeeds”: it returns the best parameters *within the family you chose*. Whether those parameters say anything true about the world is a different question, answered only by confronting the model with out-of-sample reality. These notes make that distinction concrete and measurable.

2 Tail risk: VaR and Expected Shortfall

Let L denote the loss over the risk horizon (here, one day), so that $L = -r$ for a log-return r . Fix a confidence level $\alpha \in (0, 1)$ (e.g. 0.99).

Definition 1 (Value-at-Risk). $\text{VaR}_\alpha(L)$ is the α -quantile of the loss distribution,

$$\text{VaR}_\alpha(L) = \inf\{\ell \in \mathbb{R} : \mathbb{P}(L \leq \ell) \geq \alpha\}.$$

VaR answers “how bad is a bad day, at the α threshold?” It says nothing about *how much worse* the rare day beyond the threshold can be, and it is not sub-additive — diversification can appear to *increase* it. The fix is the conditional average of the tail:

Definition 2 (Expected Shortfall).

$$\text{ES}_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 \text{VaR}_u(L) \, du = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)] \quad (\text{for continuous } L).$$

ES (a.k.a. CVaR or TVaR) is *coherent*: monotone, translation invariant, positively homogeneous, and sub-additive. It is the average loss on the worst $(1-\alpha)$ fraction of days, and it is the quantity post-2016 Basel rules put at the centre of market-risk capital. Both VaR and ES depend entirely on the *shape of the tail* — which is exactly the part of the distribution that data is sparsest about.

3 The data and its stylised facts

To study calibration against a known truth, I generate 3,000 daily log-returns (≈ 12 trading years) from a GARCH(1, 1) process with standardised Student- t innovations:

$$r_t = \mu + \sigma_t z_t, \quad \sigma_t^2 = \omega + a \varepsilon_{t-1}^2 + b \sigma_{t-1}^2, \quad z_t \sim t_\nu(\text{unit variance}),$$

with true parameters $\mu = 5 \times 10^{-4}$, $\omega = 2 \times 10^{-6}$, $a = 0.08$, $b = 0.90$, and $\nu = 4$. This process is the “reality” in the experiment. It reproduces the two stylised facts that make tail risk hard:

Fat tails. The realised series has excess kurtosis ≈ 23.7 (Gaussian = 0): extreme moves are vastly more frequent than a Normal allows.

Volatility clustering. Variance is persistent ($a + b = 0.98$): turbulent days follow turbulent days. Figure 1 shows the bursts.

A Normal QQ-plot (Figure 2) is the quickest diagnostic: if the data were Gaussian the points would lie on the line; instead the tails peel away, the signature of a heavy-tailed law.

4 Calibration: three models

I fit three models to the loss series $L = -r$ by maximum likelihood, in order of increasing tail-awareness.

Normal. Two parameters (μ, σ) , fit by the sample mean and standard deviation. Convenient, closed-form everything, and thin-tailed by construction — the null hypothesis we expect to fail.

Student- t . Adds a degrees-of-freedom parameter ν that fattens the tails; as $\nu \rightarrow \infty$ it collapses to the Normal. Fit by MLE over $(\nu, \text{loc}, \text{scale})$.

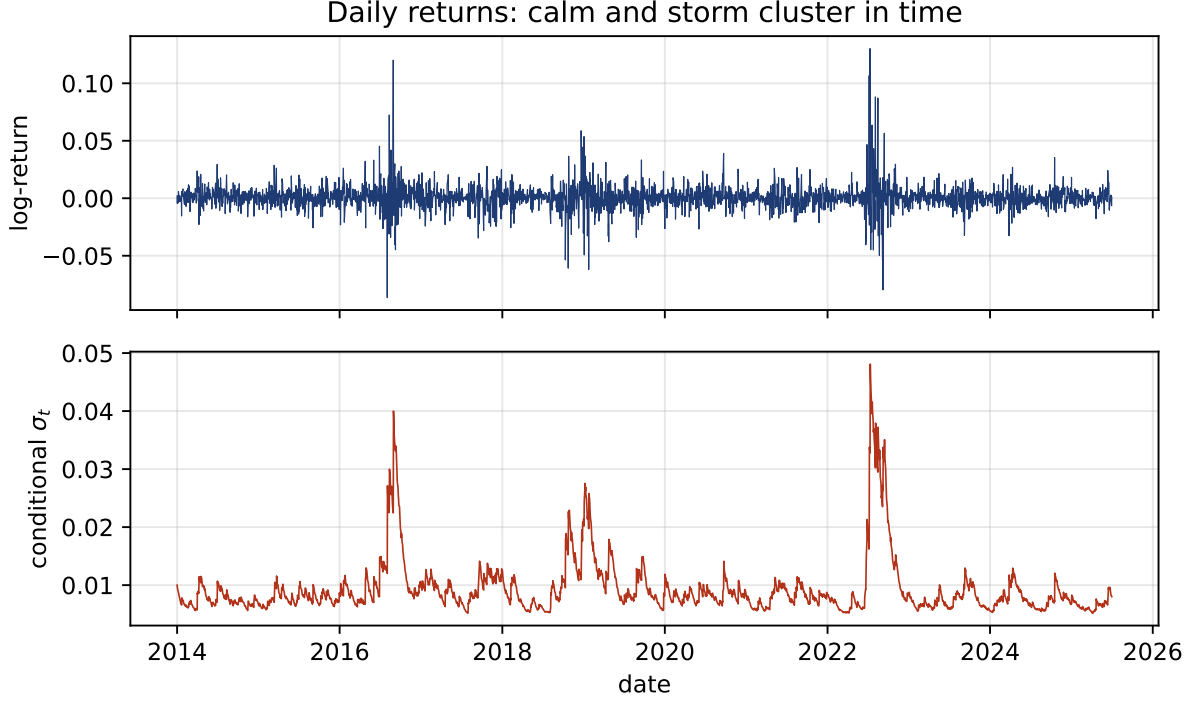


Figure 1: Top: simulated daily returns. Bottom: the latent conditional volatility σ_t . Large moves and high volatility cluster — they are not sprinkled uniformly through time.

Extreme Value Theory (Peaks-Over-Threshold). Instead of modelling the whole distribution, EVT models *only* the tail. The Pickands–Balkema–de Haan theorem says that for a high threshold u , the exceedances $L - u \mid L > u$ converge to a Generalized Pareto Distribution (GPD):

$$G_{\xi, \beta}(y) = 1 - \left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi}, \quad y > 0.$$

The shape $\xi > 0$ signals a heavy (power-law) tail. I set u at the 90% quantile, leaving 300 exceedances, and fit (ξ, β) by MLE.

Table 1: Calibrated parameters (loss space).

Model	Parameters		
Normal	$\mu = -0.00063$	$\sigma = 0.01080$	
Student- t	$\nu = 2.21$	loc= -0.00062	scale= 0.00536
GPD (POT)	$u = 0.0094$	$\xi = +0.260$	$\beta = 0.00598$

A first surprise — calibration $\neq$ truth. The true *conditional* innovation has $\nu = 4$, yet the calibrated i.i.d. Student- t reports $\nu \approx 2.21$. This is not an error; it is the central lesson appearing early. Mixing many days of *different* volatilities (the GARCH effect) produces an *unconditional* distribution with much heavier tails than any single conditional t_4 . Fitting an i.i.d. model to a clustered reality forces ν down to absorb that extra kurtosis. The number that “best fits” is faithful to the *assumed family*, not to the mechanism that generated the data. Figure 3 shows the cost of the thin-tailed assumption directly.

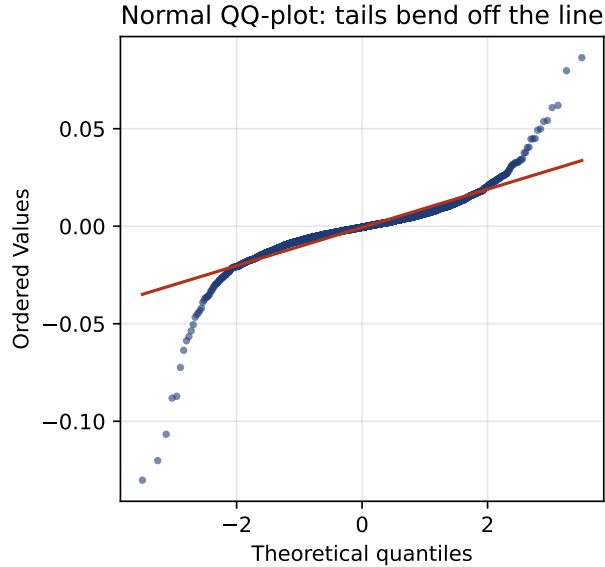


Figure 2: Normal QQ-plot of the losses. The systematic departure at both ends is the visual fingerprint of fat tails.

5 Monte Carlo simulation

For the Normal and Student- t fits, VaR and ES have closed forms. Real portfolios — options, path-dependence, many coupled risk factors — usually do not. Monte Carlo is the general method:

1. Draw N i.i.d. losses $L^{(1)}, \dots, L^{(N)}$ from the calibrated model.
2. Estimate $\widehat{\text{VaR}}_\alpha$ as the empirical α -quantile.
3. Estimate $\widehat{\text{ES}}_\alpha$ as the mean of the draws $\geq \widehat{\text{VaR}}_\alpha$.

The estimator carries *sampling noise* of order $1/\sqrt{N}$. For a quantile, the asymptotic standard error is

$$\text{s.e.}(\widehat{\text{VaR}}_\alpha) \approx \frac{1}{f(\text{VaR}_\alpha)} \sqrt{\frac{\alpha(1-\alpha)}{N}},$$

which blows up as $\alpha \rightarrow 1$ (the density f in the tail is tiny). Tail quantiles are intrinsically the hardest to simulate — precisely why variance reduction and EVT exist. Figure 4 shows convergence to the analytic value.

Table 2: Monte Carlo ($N = 200,000$) vs. closed form, calibrated Student- t . Agreement here validates the *simulation*; it says nothing about whether the model is *right*.

	$\alpha = 95\%$		$\alpha = 99\%$	
	VaR	ES	VaR	ES
Monte Carlo	0.0141	0.0279	0.0326	0.0597
Analytic	0.0141	0.0280	0.0323	0.0603

6 Calibration versus reality: backtesting

This is the heart of the matter. Each calibrated model emits a VaR. Reality — the 3,000 realised days — emits actual losses. A *breach* is a day on which the realised loss exceeds the model's VaR. At level α we expect breaches on a $(1 - \alpha)$ fraction of days. We test that with the Kupiec

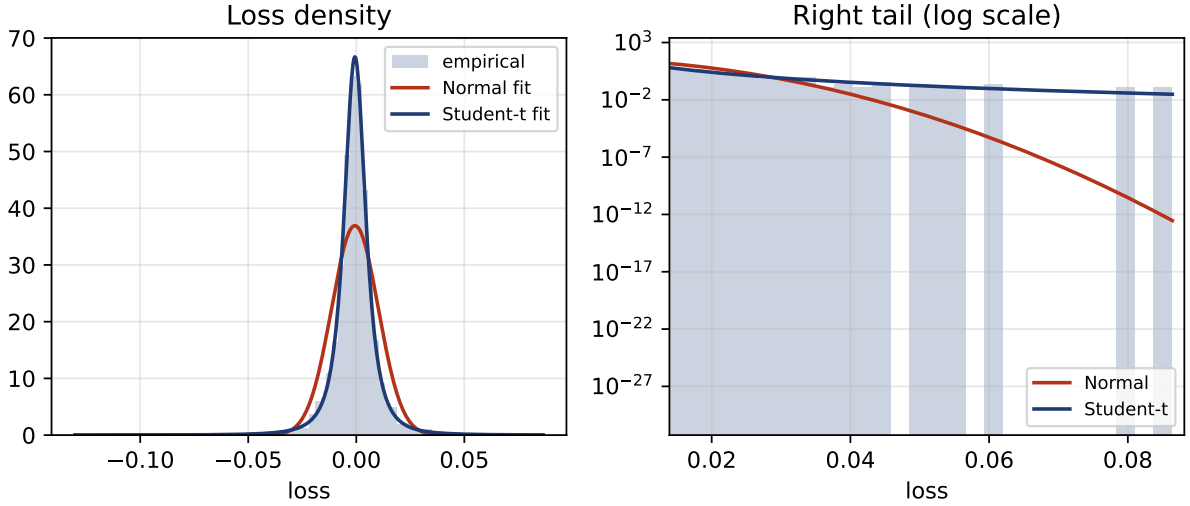


Figure 3: Left: loss histogram with the Normal and Student- t fits. Right: the same right tail on a log scale. The Normal (red) plunges to zero; the Student- t (navy) keeps mass out where the real losses live.

proportion-of-failures likelihood-ratio statistic

$$LR_{\text{POF}} = -2 \ln \frac{(1-p)^{n-x} p^x}{(1-\hat{\pi})^{n-x} \hat{\pi}^x} \xrightarrow{H_0} \chi_1^2, \quad \hat{\pi} = \frac{x}{n}, \quad p = 1 - \alpha,$$

where x is the breach count over n days. A small p -value means the observed breach rate is inconsistent with the claimed level.

Table 3: Backtest over $n = 3,000$ realised days. The Normal model is **rejected** at both levels; the fat-tailed models pass.

α	Model	VaR	Breaches	Expected	Kupiec p	Verdict
95%	Normal	0.0171	100	150.0	0.000	REJECT
	Student- t	0.0141	148	150.0	0.867	PASS
	GPD/EVT	0.0139	150	150.0	1.000	PASS
99%	Normal	0.0245	42	30.0	0.038	REJECT
	Student- t	0.0323	23	30.0	0.180	PASS
	GPD/EVT	0.0283	29	30.0	0.854	PASS

Read the Normal row carefully, because it shows *two opposite failures of the same model*:

- At 95% it is *too conservative* (100 breaches vs. 150): its wide “body” overstates ordinary risk and wastes capital.
- At 99% it is *too reckless* (42 breaches vs. 30): its thin tail understates extreme risk — the failure that bankrupts you.

A single mis-specified family is simultaneously charging too much for calm and too little for catastrophe. Figure 5 marks where reality broke through the 99% lines.

6.1 Counting is not enough: the independence test

Kupiec asks *how many* breaches occurred. It is blind to *when*. A model that breaches 150 times in 3,000 days passes it perfectly — even if all 150 fell in three bad months. That matters: a desk

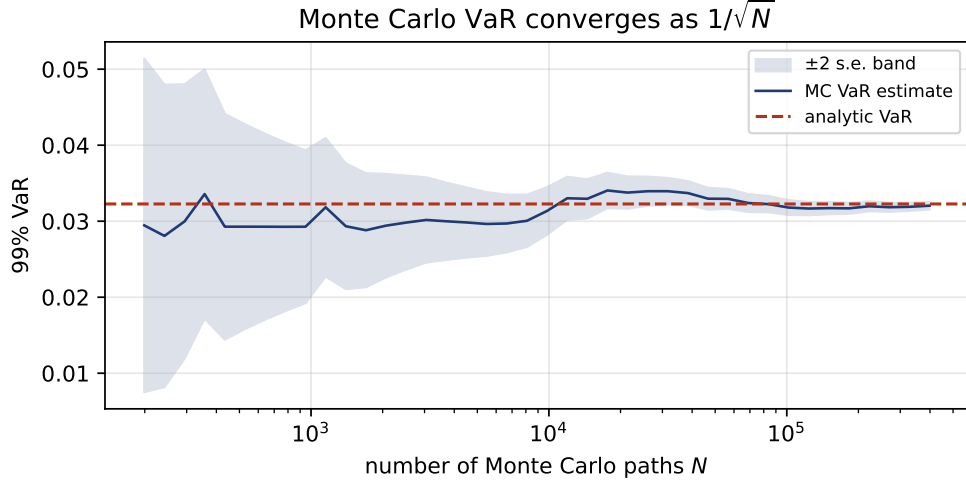


Figure 4: Monte Carlo 99% VaR versus number of paths N , with a ± 2 standard-error band, converging to the closed-form value (dashed).

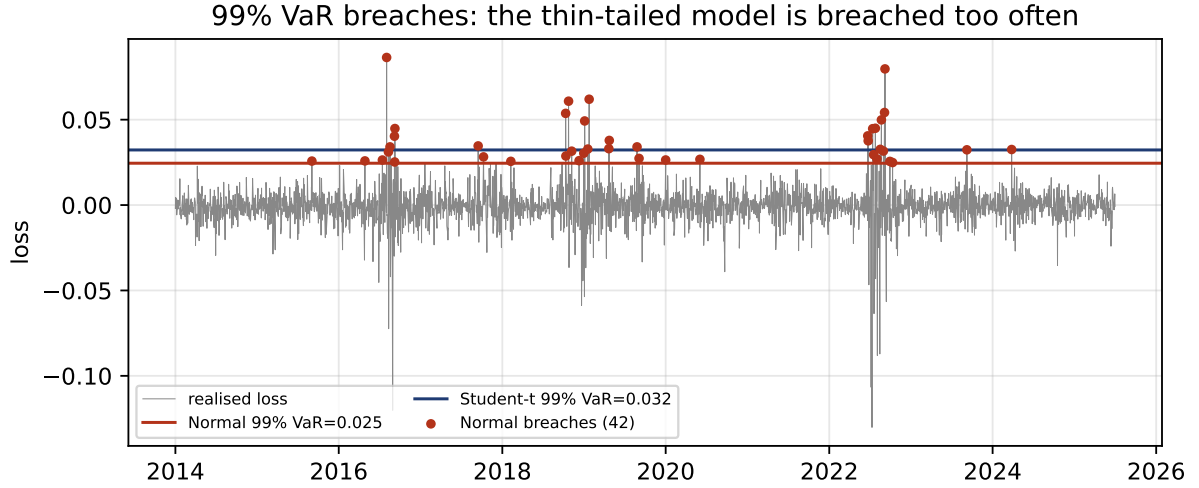


Figure 5: Realised losses with each model's 99% VaR. Red dots are days that breached the Normal VaR — visibly clustered, another reminder that breaches arrive together, not independently.

that loses its VaR several days running is in far more trouble than one that loses it on scattered days, and a GARCH reality produces exactly that pattern.

Christoffersen's test fits a two-state Markov chain to the 0/1 breach sequence. Let n_{ij} count days in state j following a day in state i ($1 = \text{breach}$), and set

$$\hat{\pi}_{01} = \frac{n_{01}}{n_{00} + n_{01}}, \quad \hat{\pi}_{11} = \frac{n_{11}}{n_{10} + n_{11}}, \quad \hat{\pi} = \frac{n_{01} + n_{11}}{n_{00} + n_{01} + n_{10} + n_{11}}.$$

$\hat{\pi}_{11}$ is the breach rate on the day *after* a breach. Under independence it equals $\hat{\pi}_{01}$, and

$$LR_{\text{ind}} = -2 \ln \frac{(1 - \hat{\pi})^{n_{00} + n_{10}} \hat{\pi}^{n_{01} + n_{11}}}{(1 - \hat{\pi}_{01})^{n_{00}} \hat{\pi}_{01}^{n_{01}} (1 - \hat{\pi}_{11})^{n_{10}} \hat{\pi}_{11}^{n_{11}}} \xrightarrow{H_0} \chi_1^2.$$

The sum $LR_{\text{cc}} = LR_{\text{POF}} + LR_{\text{ind}} \sim \chi_2^2$ tests count and timing jointly (“conditional coverage”). At 95%, a breach triples the chance of another the next day. Fixing the tail *shape* (Student- t , EVT) fixed the *count*; it did nothing for the *timing*, because every model here is unconditional

Table 4: Independence of breaches. At 95% **every** model is rejected — including the two that passed Kupiec.

α	Model	Kupiec p	$\hat{\pi}_{11}$	$\hat{\pi}_{01}$	Christoffersen p
95%	Normal	0.000	11.0%	3.1%	0.000
	Student- t	0.867	14.2%	4.5%	0.000
	GPD/EVT	1.000	14.0%	4.5%	0.000
99%	Normal	0.038	7.1%	1.3%	0.022
	Student- t	0.180	0.0%	0.8%	0.551
	GPD/EVT	0.854	3.4%	0.9%	0.286

— one VaR number for calm and storm alike. The 99% passes deserve suspicion rather than comfort: with only 23–29 breaches, just 0–1 of them fall on consecutive days, so the test has almost no power and the χ^2 approximation is rough. A pass there means “not enough evidence of clustering,” not “independent.”

7 The fix: a VaR that moves with volatility

Every model so far answers “how fat is the tail?” and ignores “how stormy is today?”. GARCH-filtered EVT (McNeil and Frey, 2000) answers the two questions separately.

1. **GARCH(1,1) for the storm.** Fit $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$ by Gaussian quasi-maximum likelihood. The Normal likelihood still gives consistent GARCH parameters when the true shocks are fat-tailed. It is only wrong about the tail, which step 2 handles instead. σ_t uses information up to day $t - 1$ only.
2. **EVT for the tail.** Standardise, $z_t = (r_t - \mu)/\sigma_t$. With the clustering divided out, the z_t are close to i.i.d., so fitting a POT–GPD to their loss tail is now well posed. It gives a fixed tail quantile $q_z(p)$.
3. **Recombine.** $\text{VaR}_t(p) = -\mu + \sigma_t q_z(p)$: a fixed tail shape scaled by a moving volatility.

Table 5: GARCH-EVT against the best unconditional model. It is the only model that passes both tests at both levels.

α	Model	mean VaR	Breaches	Kupiec p	$\hat{\pi}_{11}$	$\hat{\pi}_{01}$	Christoffersen p
95%	Student- t	0.0141	148	0.867	14.2%	4.5%	0.000
	GARCH-EVT	0.0139	151	0.933	6.0%	5.0%	0.603
99%	Student- t	0.0323	23	0.180	0.0%	0.8%	0.551
	GARCH-EVT	0.0247	28	0.711	0.0%	0.9%	0.468

At 95%, the day after a breach now carries a 6.0% breach rate against a 5.0% base, where the static models showed 14% against 4.5%. The clustering is gone (Figure 6). The capital story is just as important. At 99%, GARCH-EVT holds *less* VaR on average than the Student- t (0.0247 against 0.0323) and still delivers the right breach count. It holds less in calm periods and more in storms, where the static model holds the same amount everywhere and so over-reserves in calm and under-reserves in crisis.

Checks that the pieces behave.

- *GARCH recovers the truth.* $\hat{\alpha} = 0.097$, $\hat{\beta} = 0.888$, against a true 0.08 and 0.90. When the model family matches reality, calibration finds reality. Contrast the unconditional $\hat{\nu} = 2.21$ against a true 4.

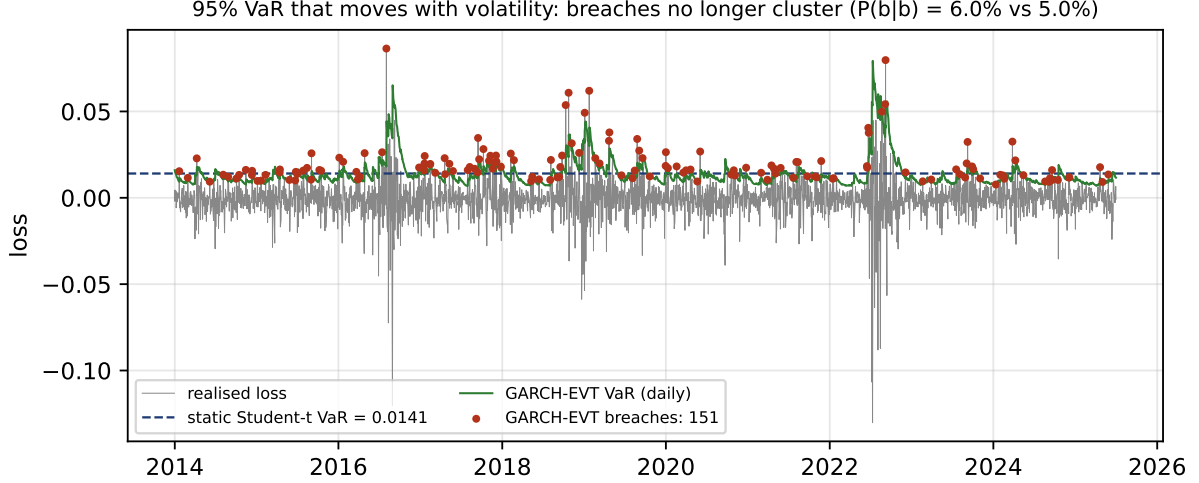


Figure 6: GARCH-EVT 95% VaR (green) rises into the 2016 and 2022 storms and relaxes after them. Its breaches are spread across the sample instead of bunched into the volatile spells.

- *Filtering removes the clustering.* Excess kurtosis falls from 23.7 on raw returns to 4.5 on the residuals. What remains is the genuine fat tail of the t_4 shocks.
- *The residual tail index is biased low, as expected.* The fit gives $\hat{\xi} = 0.12$, where a t_4 tail has $\xi = 1/4$. This is not a bug. A Student- t tail is Pareto only asymptotically, so a GPD fitted above a moderate threshold underestimates ξ . On two million clean t_4 draws the same procedure gives $\hat{\xi} \approx 0.17$ at the 90th percentile, 0.22 at the 99th and 0.30 at the 99.9th. With 300 exceedances (standard error ≈ 0.065), 0.12 is within about one standard error of 0.17.

A caveat on everything above. This backtest is *in-sample*: each model is scored on the same 3,000 days it was fitted to, which flatters all of them. The GPD's perfect 150/150 at 95% is close to automatic, since its threshold is this sample's 90th percentile. For GARCH-EVT, each day's σ_t uses only past returns, but the parameters were estimated on the full sample. A rolling out-of-sample backtest is the honest next step.

8 What this says about calibration and reality

1. **Fitting the body is not fitting the tail.** The Normal nails the mean and variance and still fails where it matters. Goodness-of-fit in the centre is no evidence about the extremes.
2. **Calibration recovers the model, not the truth.** The fitted $\nu = 2.21$ is the right answer to the wrong question; the true conditional ν was 4. An i.i.d. lens cannot see a clustered world, so it relabels persistence as fatter tails.
3. **Simulation accuracy $\neq$ model accuracy.** Table 2 shows Monte Carlo reproducing the analytic VaR to four decimals. That only proves the *arithmetic* is right. Backtesting (Table 3) is what tests the *model*.
4. **Estimate ES, and prefer EVT in the tail.** ES integrates the whole tail and is coherent; EVT models the tail on its own terms and here matched the breach *count* most closely (5.00% and 0.97%) — though in-sample, and not the breach *timing*.
5. **Passing Kupiec is not the number being right.** Breaches cluster (Table 4): every model fails independence at 95%. A fatter tail cannot fix that. A *conditional* VaR that moves with current volatility can: GARCH-EVT (Table 5) passes both tests at both levels.

6. **Get the model family right, and calibration finds the truth.** The same MLE machinery that returned $\nu = 2.21$ for a static t recovers the true GARCH parameters to within 0.02 once the model can express clustering. Most of calibration's success or failure is decided by the choice of model family.

The map is not the territory. A calibrated model is a map drawn from the ground we have already walked. Tail risk lives in the ground we have not. The discipline is to keep asking how the map could be wrong — by backtesting, by stressing parameters, and by choosing families (Student- t , EVT) honest about how heavy reality's tails can be.

A Reproducing these notes

From `src/` run, in order:

```
python3 generate_data.py    # writes data/returns.csv (the "reality")
python3 calibrate.py        # fits Normal / Student-t / GPD -> results.json
python3 conditional.py      # GARCH(1,1) QMLE + EVT on residuals -> daily VaR
python3 simulate.py         # Monte Carlo VaR & ES
python3 backtest.py         # Kupiec + Christoffersen vs realised losses
python3 plots.py            # regenerates every figure in this document
```

All randomness is seeded in `config.py`, so the numbers quoted here reproduce exactly. Change `TRUE_NU`, `TRUE_BETA`, or the confidence levels there to run your own calibration-versus-reality experiments.